GATE ME 90 Days Challenge

Day 61/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Manufacturing	**Machine Tools, NTM & Additive**	**40 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on lathe operations and machining time, thread cutting, taper turning, drilling, shaping and planing, milling and indexing, jigs and fixtures (3-2-1 location, bushes, diamond pins), non-traditional machining (EDM relaxation circuit, ECM by Faraday's law, USM, AJM, LBM) and additive manufacturing processes and build time. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

During thread cutting on a centre lathe, the carriage is moved by

(A) the feed rod through the apron gearing

(B) engaging the split (half) nut with the lead screw, so that the carriage advances one lead per revolution of the work

(C) hand, using the carriage hand-wheel

(D) the cross-slide screw

Q2 NAT

A bar of 60 mm diameter is reduced to 52 mm diameter over a length of 150 mm in two passes of 2 mm depth each. The cutting speed is kept at 120 m/min, calculated on the diameter at the start of each pass. The approach is 5 mm and the feed is 0.25 mm/rev. The total machining time (in min, round off to two decimal places) is _____

Q3 MCQ

The essential difference between a jig and a fixture is that

(A) a fixture is used only for drilling

(B) a jig is always heavier and is bolted to the machine table

(C) a jig is used only on lathes

(D) a jig holds and locates the work and also guides the cutting tool (e.g. through drill bushes), whereas a fixture only holds and locates the work

Q4 NAT

In an RC relaxation circuit for EDM, the supply voltage is 100 V, R = 40 Ω and C = 20 μF. The discharge across the gap occurs when the capacitor voltage reaches 70 V. Neglecting the discharge time, the spark frequency (in Hz, round off to the nearest whole number) is _____

Q5 MCQ

In electrochemical machining, the tool shows practically no wear because

(A) the tool is made of diamond

(B) the tool is the cathode, where only hydrogen gas is evolved and no metal is dissolved

(C) the electrolyte is a dielectric

(D) the tool is cooled by the electrolyte

Q6 NAT

Aluminium (atomic weight 27, valency 3, density 2.7 g/cm³) is machined by ECM at 500 A. Taking Faraday's constant as 96 500 C/mol and 100 % efficiency, the volumetric material removal rate (in mm³/s, round off to two decimal places) is _____

Q7 MCQ

Typical operating values of tool vibration in ultrasonic machining are

(A) frequency about 20 kHz and amplitude about 15–50 µm

(B) frequency about 50 Hz and amplitude about 1 mm

(C) frequency about 2 MHz and amplitude about 1 nm

(D) frequency about 20 kHz and amplitude about 5 mm

Q8 NAT

A dividing head has a 40 : 1 worm gear, and index plates with hole circles of 15, 16, 17, 18, 19, 20 / 21, 23, 27, 29, 31, 33 / 37, 39, 41, 43, 47, 49. To mill 52 equally spaced slots by simple indexing using the 39-hole circle, the number of holes the crank must move for each slot is _____

Q9 MCQ

Match the additive manufacturing process (Column I) with its category (Column II).
P. Stereolithography (SLA) Q. Fused deposition modelling (FDM) R. Selective laser sintering (SLS) S. 3D printing with a liquid binder
1. Powder bed fusion 2. Binder jetting 3. Vat photopolymerisation 4. Material extrusion

(A) P-1, Q-4, R-3, S-2

(B) P-3, Q-2, R-1, S-4

(C) P-4, Q-3, R-1, S-2

(D) P-3, Q-4, R-1, S-2

Q10 NAT

A part 36 mm high is built by stereolithography with a layer thickness of 0.1 mm. Each layer takes 20 s for laser scanning and 10 s for recoating. The build time (in min) is _____

Q11 MCQ

Abrasive jet machining is NOT well suited for

(A) cutting thin sections of glass

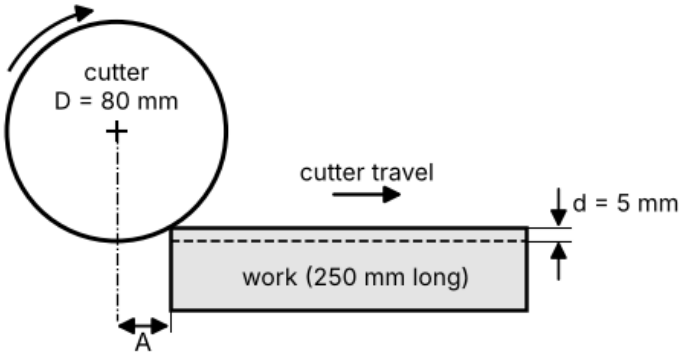
(B) deburring and frosting of hard brittle parts

(C) machining soft, ductile materials such as lead or soft aluminium, where the abrasive embeds in the surface

(D) drilling small holes in ceramics

Q12 NAT

A surface 250 mm long is slab milled with a cutter of 80 mm diameter and 6 teeth at 200 rpm, with a feed of 0.15 mm per tooth and a depth of cut of 5 mm, as shown. Taking the approach as $\sqrt{d(D-d)}$ and neglecting over-travel, the machining time (in min, round off to two decimal places) is _____



Q13 MCQ

In a planing machine, as distinct from a shaper,

(A) the work, mounted on the table, reciprocates under a stationary tool, and the tool is fed between strokes

(B) the tool reciprocates and the work is fed

(C) the work rotates and the tool is fed axially

(D) the tool rotates and the work is fed

Q14 NAT

A hole of 0.4 mm diameter is to be drilled through a 1.5 mm thick steel sheet by a pulsed laser. Each pulse has an energy of 2 J, of which 40 % is absorbed, and the energy needed to vaporise steel is 60 J/mm³. The minimum number of pulses needed is _____

Q15 MCQ

In the 3-2-1 principle of location for a prismatic workpiece,

(A) three clamps, two locators and one support are used

(B) the work is held by three jaws, two centres and one stop

(C) only three locators are needed for complete location

(D) six locators are used: three on the primary (base) plane, two on the secondary plane and one on the tertiary plane, which together with the clamps restrain all 12 degrees of freedom

Answer Key

Q	Type	Answer
Q1	MCQ	(B) engaging the split (half) nut with the lead screw, so that the carriage advances one lead per revolution of the work
Q2	NAT	1.88 (accept 1.87 to 1.90)
Q3	MCQ	(D) a jig holds and locates the work and also guides the cutting tool (e.g. through drill bushes), whereas a fixture only holds and locates the work
Q4	NAT	1038 (accept 1030 to 1046)
Q5	MCQ	(B) the tool is the cathode, where only hydrogen gas is evolved and no metal is dissolved
Q6	NAT	17.27 (accept 17.15 to 17.39)
Q7	MCQ	(A) frequency about 20 kHz and amplitude about 15–50 μm
Q8	NAT	30
Q9	MCQ	(D) P-3, Q-4, R-1, S-2
Q10	NAT	180
Q11	MCQ	(C) machining soft, ductile materials such as lead or soft aluminium, where the abrasive embeds in the surface
Q12	NAT	1.50 (accept 1.49 to 1.51)
Q13	MCQ	(A) the work, mounted on the table, reciprocates under a stationary tool, and the tool is fed between strokes
Q14	NAT	15
Q15	MCQ	(D) six locators are used: three on the primary (base) plane, two on the secondary plane and one on the tertiary plane, which together with the clamps restrain all 12 degrees of freedom

Detailed Solutions

Q1 Answer: (B) engaging the split (half) nut with the lead screw, so that the carriage advances one lead per revolution of the work

Concept: A thread needs an exact, positive ratio between work rotation and tool travel.

The lead screw, driven from the spindle through change gears, moves the carriage when the half nut is closed on it: **(B)**.

The feed rod (with friction clutches in the apron) is used for ordinary turning feeds, where slip does not matter.

Exam tip: The thread-chasing dial tells the operator when to re-engage the half nut so the tool falls into the same groove.

Q2 **Answer: 1.88 (accept 1.87 to 1.90)**

Concept: t = L/(fN) per pass, with N = 1000V/(πD) recalculated for each pass.

Pass 1: D = 60 mm, N = 120 000/(π × 60) = 636.6 rpm; t_1 = 155/(0.25 × 636.6) = 0.974 min

Pass 2: D = 56 mm, N = 120 000/(π × 56) = 682.1 rpm; t_2 = 155/(0.25 × 682.1) = 0.909 min

Total = **1.88 min**

Exam tip: Each pass of depth 2 mm reduces the diameter by 4 mm.

Q3 **Answer: (D) a jig holds and locates the work and also guides the cutting tool (e.g. through drill bushes), whereas a fixture only holds and locates the work**

Concept: Both locate and clamp the workpiece in a fixed position.

A jig additionally guides the tool, typically with hardened bushes for drilling, reaming and tapping. A fixture is fixed to the machine table and the tool is set with setting blocks or feeler gauges, as in milling, turning and grinding: **(D)**.

Exam tip: Drill jigs are usually not clamped to the table; milling fixtures are.

Q4 **Answer: 1038 (accept 1030 to 1046)**

Concept: The capacitor charges as $V_c = V_s(1 - e^{-t/RC})$; the cycle time is the time to reach the discharge voltage.

$RC = 40 \times 20 \times 10^{-6} = 8 \times 10^{-4}$ s

$t = RC \ln[V_s/(V_s - V_d)] = 8 \times 10^{-4} \times \ln(100/30) = 9.632 \times 10^{-4}$ s

f = 1/t = **1038 Hz**

Exam tip: The energy per spark is $\frac{1}{2}CV_d^2$ = 0.049 J here; a higher V_d gives bigger sparks but a lower frequency.

Q5 **Answer: (B) the tool is the cathode, where only hydrogen gas is evolved and no metal is dissolved**

Concept: In electrolysis metal dissolves only at the anode.

The work is made the anode and dissolves; at the tool (cathode) the reaction is the release of hydrogen gas (and hydroxide ions), so the tool shape is kept: **(B)**.

Exam tip: The electrolyte (e.g. NaCl or $NaNO_3$ solution) is a good conductor, unlike the dielectric used in EDM.

Q6 **Answer: 17.27 (accept 17.15 to 17.39)**

Concept: Faraday's law: mass removal rate = IA/(nF).

$\dot{m} = 500 \times 27/(3 \times 96\,500) = 0.04663$ g/s

Q = 0.04663/2.7 = 0.01727 cm³/s = **17.27 mm³/s**

Exam tip: The MRR does not depend on the hardness of the work, which is why ECM suits hard alloys.

Q7 **Answer: (A) frequency about 20 kHz and amplitude about 15–50 µm**

Concept: The magnetostrictive or piezoelectric transducer works just above the audible range, and a horn (concentrator) magnifies the amplitude.

Frequency ≈ 19–25 kHz and amplitude ≈ 15–50 µm: **(A)**.

Exam tip: In Shaw's model, the MRR increases with amplitude, frequency and grain size, and is highest for brittle work materials.

Q8 **Answer: 30**

Concept: In simple indexing, crank turns per division = 40/N.

40/52 = 10/13 (less than one turn)

10/13 = 30/39, so the crank moves **30 holes** on the 39-hole circle (no full turns).

Exam tip: The sector arms are set to span 30 spaces (31 holes) so that the count cannot go wrong.

Q9 **Answer: (D) P–3, Q–4, R–1, S–2**

Concept: ISO/ASTM 52900 groups AM processes by how each layer is made.

SLA cures liquid resin in a vat (vat photopolymerisation); FDM extrudes a molten filament (material extrusion); SLS fuses powder with a laser (powder bed fusion); binder jetting glues powder with a liquid binder: **(D)**.

Exam tip: SLS and binder jetting need no support structures because the loose powder supports the part; SLA and FDM need supports for overhangs.

Q10 **Answer: 180**

Concept: Build time = number of layers × time per layer.

Layers = 36/0.1 = 360

Time = 360 × (20 + 10) = 10 800 s = **180 min**

Exam tip: Halving the layer thickness improves the surface (smaller stair steps) but doubles the number of layers and so roughly doubles the build time.

Q11 **Answer: (C) machining soft, ductile materials such as lead or soft aluminium, where the abrasive embeds in the surface**

Concept: In AJM a high-velocity stream of fine abrasive in a gas jet erodes the work by brittle fracture.

Ductile materials deform instead of chipping, and the grains tend to embed in soft surfaces, so the MRR is very low: **(C)**.

Exam tip: Typical abrasives are Al_2O_3 and SiC of 10–50 µm, at gas pressures of a few bar and jet velocities of 150–300 m/s.

Q12 **Answer: 1.50 (accept 1.49 to 1.51)**

Concept: The cutter reaches full depth only after its centre has moved the approach A = √(d(D − d)) from the point of first contact.

A = √(5 × 75) = 19.36 mm

Table feed = $f_z zN$ = 0.15 × 6 × 200 = 180 mm/min

t = (250 + 19.36)/180 = **1.50 min**

Exam tip: In face milling with the cutter central, the approach is D/2 instead.

Q13 **Answer: (A) the work, mounted on the table, reciprocates under a stationary tool, and the tool is fed between strokes**

Concept: Both produce flat surfaces with single-point tools in straight strokes.

In a shaper the ram carries the tool over a stationary work; in a planer the heavy table carries the large work past the tool: **(A)**.

Exam tip: Planers are used for large, heavy jobs (machine beds); shapers for small jobs.

Q14 **Answer: 15**

Concept: Pulses needed = energy to remove the hole volume/absorbed energy per pulse (rounded up).

Volume = $(\pi/4) \times 0.4^2 \times 1.5 = 0.1885\ mm^3$

Energy = $0.1885 \times 60 = 11.31$ J; absorbed per pulse = $0.4 \times 2 = 0.8$ J

Pulses = 11.31/0.8 = 14.14, so **15 pulses**

Exam tip: Always round up the number of pulses (or passes); 14 pulses would leave the hole incomplete.

Q15 **Answer: (D) six locators are used: three on the primary (base) plane, two on the secondary plane and one on the tertiary plane, which together with the clamps restrain all 12 degrees of freedom**

Concept: A free body has 12 degrees of freedom (±translation and ±rotation about three axes).

Three points on the base restrain 5 (−Z and rotations about X and Y, both senses), two on a side face restrain 3 more (−X and both senses of rotation about Z), and one on the end face restrains 1 more (−Y): 9 by the 6 locators.

The clamps restrain the remaining 3 (+X, +Y, +Z): **(D)**.

Exam tip: The three base locators should be spread as widely as possible for a stable seating.

What next?

Day 61 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 62/90: Manufacturing, Metrology & CIM. Comment **GATE90** on our reel to get all links.
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